

Cost Estimates

Project: Aura Promptly
Version: 1.1
Date: 2026-05-14
Status: Baselined (updated post-review)

Change Log

Version	Date	Changes
1.0	2026-05-14	Initial baseline
1.1	2026-05-14	Section 2.3 updated: Aurora pgvector replaces OpenSearch Serverless (~90% cost reduction); Section 3 scenarios recalculated; Section 4.1 toggle variables updated; Section 4.3 recreate runbook updated with KB sync poll-and-wait

All prices are approximate, based on AWS us-east-1 pricing as of May 2026. Actual costs depend on usage volume. Use Infracost (`infracost breakdown --path ./infra`) for current exact estimates.

1. Cost Philosophy

The PoC infrastructure is designed to be **destroyed when not in use** and **recreated in under 30 minutes**. The cost strategy is:

- Pay only when actively developing or demoing
- Aurora Serverless scales to 0 ACUs when idle — the KB vector store costs nothing between sessions
- Enable/disable optional components (WAF, Redis, Grafana export) via Terraform variables
- The two highest-cost items in v1.0 (OpenSearch Serverless ~\$172/month, NAT Gateway ~\$32/month) are now either eliminated or toggleable

2. Component Cost Breakdown

2.1 Hub Account (AWS)

Component	Service	Pricing Model	Always-on monthly cost	Active (8h/day, 20 days)
LiteLLM Proxy	ECS Fargate (0.25 vCPU, 0.5 GB)	Per task-hour	~\$3/month	~\$3.50/month
LiteLLM DB	Aurora Serverless (PostgreSQL)	Per ACU-hour (min 0.5 ACU, auto-pauses)	~\$4.50/month	~\$8/month

Component	Service	Pricing Model	Always-on monthly cost	Active (8h/day, 20 days)
Response Cache	ElastiCache Redis (cache.t4g.micro)	Per hour	~\$12/month	~\$12/month
API Gateway	REST API	Per million requests	~\$0 (PoC volume)	<\$1/month
WAF	AWS WAF WebACL	\$5/WebACL + \$1/million requests	\$5/month	\$5/month
Cognito	User Pool	Free up to 50k MAUs	\$0	\$0
Secrets Manager	Secrets	\$0.40/secret/month	~\$2/month	~\$2/month
S3	Storage (config, KB docs)	\$0.023/GB	<\$1/month	<\$1/month
NAT Gateway	Per hour + data	\$0.045/hour + \$0.045/GB	~\$32/month (always on)	~\$32/month
VPC Endpoints	Per endpoint-hour	\$0.01/endpoint/hour (7 endpoints)	~\$50/month	~\$50/month
CloudWatch Logs	Ingestion + storage	\$0.50/GB ingested	~\$2/month	~\$3/month
OTel Collector	ECS Fargate (0.25 vCPU, 0.5 GB)	Per task-hour	~\$3/month	~\$3.50/month

Note on NAT vs VPC Endpoints: VPC endpoints cost ~\$50/month (7 endpoints) but NAT Gateway costs ~\$32/month. For small traffic volumes, NAT is cheaper. For high Bedrock traffic, VPC endpoints save on data transfer costs. The Terraform toggle `enable_vpc_endpoints` lets you switch.

Hub base cost (always-on, no AI services running): ~\$65–85/month

To reduce hub cost during idle periods:

- Set ECS desired count to 0 (Fargate tasks stop immediately)
- Aurora auto-pauses after 5 minutes — effectively ~\$0 when idle

Hub cost when fully idle (ECS stopped, Aurora paused): ~\$15–20/month (NAT GW, VPC endpoints, Secrets Manager, S3)

2.2 AI Services (On-Demand, Usage-Based)

Service	Unit Cost	PoC Estimate (100 requests/day)
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Service	Unit Cost	PoC Estimate (100 requests/day)
Bedrock Claude Haiku 3	\$0.00025/1K input + \$0.00125/1K output	~\$0.30/day (~500 input, 200 output tokens avg)
Bedrock Titan Embeddings v2	\$0.00002/1K tokens	~\$0.01/day (RAG indexing)
Bedrock Guardrails	\$0.75/1K text units	~\$0.08/day
OpenAI GPT-4o-mini	\$0.00015/1K input + \$0.0006/1K output	~\$0.10/day (fallback only)
Vertex AI Gemini Flash	\$0.000075/1K input + \$0.0003/1K output	~\$0.05/day (if routed)

Total AI usage cost (100 requests/day active session): ~\$0.54/day = ~\$11/month

2.3 Knowledge Base / RAG

Component	Cost	Notes
Aurora Serverless (pgvector schema)	~\$0 when idle (scale-to-zero)	Shared cluster with LiteLLM DB — no extra cluster cost
Aurora Serverless (active, 0.5 ACU)	~\$0.06/ACU-hour × 0.5 = ~\$0.03/hour	Only bills when queries are running
Aurora Serverless (8h/day, 20 days)	~\$5–10/month	Realistic active development cost
S3 (document storage)	\$0.023/GB	<\$1/month for PoC docs
Bedrock KB ingestion	Included in embedding cost	~\$0.01 per sync

Cost comparison vs v1.0: OpenSearch Serverless cost \$172/month (always-on) or \$58/month (8h/day). Aurora pgvector costs ~\$5–10/month active and **\$0 when idle**. This is a ~90% reduction and eliminates the need to destroy the KB between sessions.

Production note: At high QPS (>500 req/s), migrate to OpenSearch Serverless for better vector search performance. The Bedrock KB API is identical — only the backing store changes.

2.4 Spoke Accounts

Spoke	Service	Monthly Cost
Azure (Container Apps)	Consumption plan (serverless)	~\$0–3/month (free tier covers PoC)
AWS (ECS Fargate or Lambda)	Per request / per task-hour	~\$2–5/month
GCP (Cloud Run)	Per request (serverless)	~\$0–2/month (free tier covers PoC)

Total spokes: ~\$5–10/month

2.5 Tooling (Free Tiers)

Tool	Tier	Cost
Terraform Cloud	Free (1 user, 500 resources)	\$0
Grafana Cloud	Free (10k metrics, 50GB logs)	\$0
GitHub Actions	Free (2,000 min/month private repos)	\$0

3. Cost Scenarios

Scenario A: Actively Developing (8 hours/day, 5 days/week)

All components running, moderate AI usage:

Category	Monthly Cost
Hub infrastructure (ECS, Aurora, Redis, WAF, NAT)	~\$65
Aurora pgvector KB (8h/day × 20 days, 0.5 ACU)	~\$8
AI usage (100 req/day)	~\$11
Spokes	~\$8
Total	~\$92/month

v1.0 comparison: Was ~\$152/month. Saving: ~\$60/month from eliminating OpenSearch Serverless.

Scenario B: Demo Day Only (2 hours, once)

Category	Cost
Hub (2 hours)	~\$0.50
Aurora pgvector KB (2 hours, 0.5 ACU)	~\$0.06
AI usage (200 requests during demo)	~\$0.11
Spokes (2 hours)	~\$0.10
Total	~\$0.77 per demo session

v1.0 comparison: Was ~\$1.67 per session. OpenSearch alone was \$0.96 for 2 hours.

Scenario C: Idle (hub running, KB not actively queried)

Aurora scales to 0 ACUs after 5 minutes of inactivity.

Category	Monthly Cost
NAT Gateway (always-on)	~\$32
ECS Fargate (LiteLLM, 1 task)	~\$3

Category	Monthly Cost
Aurora Serverless (paused, 0 ACU)	~\$0
WAF	~\$5
Secrets Manager, S3, CloudWatch	~\$5
Total	~\$45/month

Scenario D: Fully Destroyed (no active infrastructure)

Cost: \$0/day (Terraform state in Terraform Cloud, code in GitHub — no AWS resources running)

4. Cost Control Mechanisms

4.1 Terraform Toggle Variables

```
# infra/hub/variables.tf
variable "enable_waf"           { default = true } # saves $5/month
if false
variable "enable_redis_cache"   { default = true } # saves $12/month
if false
variable "enable_vpc_endpoints" { default = false } # saves $50/month
if false (use NAT instead)
variable "enable_knowledge_base" { default = false } # drops Aurora
pgvector schema when false
variable "enable_grafana_export" { default = true } # saves ~$0 (free
tier) but stops data egress
variable "enable_otel_collector" { default = true } # saves ~$3/month
if false
```

Minimum viable PoC cost (WAF off, KB off, Redis off, VPC endpoints off, Grafana off):

~\$20–30/month when actively running, ~\$5/month idle.

4.2 Destroy Runbook

```
# Step 1: Destroy spokes first (to remove API keys and IAM roles
referencing hub)
terraform -chdir=infra/gcp-spoke destroy -auto-approve
terraform -chdir=infra/azure-spoke destroy -auto-approve
terraform -chdir=infra/aws-spoke destroy -auto-approve

# Step 2: Destroy hub
terraform -chdir=infra/hub destroy -auto-approve

# Verification: no resources should remain in AWS account
aws resourcegroupstaggingapi get-resources \
```

```
--tag-filters Key=project,Values=aura-promptly \
--query 'ResourceTagMappingList[*].ResourceARN'
```

4.3 Recreate Runbook

```
# Step 1: Deploy hub
terraform -chdir=infra/hub apply -auto-approve

# Step 2: Upload KB documents to S3
aws s3 sync ./data/kb-documents/ s3://$(terraform -chdir=infra/hub output
--raw kb_bucket_name)/

# Step 3: Trigger Knowledge Base sync
JOB_ID=$(aws bedrock-agent start-ingestion-job \
  --knowledge-base-id $(terraform -chdir=infra/hub output --raw kb_id) \
  --data-source-id $(terraform -chdir=infra/hub output --raw
kb_data_source_id) \
  --query 'ingestionJob.ingestionJobId' --output text)

# Step 4: Poll until sync completes (MUST complete before spokes start
serving RAG queries)
echo "Waiting for KB sync to complete (job: $JOB_ID)..."
while true; do
  STATUS=$(aws bedrock-agent get-ingestion-job \
    --knowledge-base-id $(terraform -chdir=infra/hub output --raw kb_id) \
    --data-source-id $(terraform -chdir=infra/hub output --raw
kb_data_source_id) \
    --ingestion-job-id $JOB_ID \
    --query 'ingestionJob.status' --output text)
  echo "  Status: $STATUS"
  [ "$STATUS" = "COMPLETE" ] && echo "KB sync complete." && break
  [ "$STATUS" = "FAILED" ]    && echo "KB sync FAILED - check CloudWatch
logs." && exit 1
  sleep 15
done

# Step 5: Deploy spokes (safe to start now - KB is ready)
terraform -chdir=infra/aws-spoke apply -auto-approve
terraform -chdir=infra/azure-spoke apply -auto-approve
terraform -chdir=infra/gcp-spoke apply -auto-approve
```

Why the poll step matters: Without it, the Azure RAG chatbot returns empty results immediately after a fresh deploy. The sync typically takes 1–5 minutes for small document sets — well within the NFR-02 <30 minute recreate target.

4.4 Infracost Integration

Every `terraform plan` in CI/CD will include an Infracost cost estimate:

```
# .github/workflows/terraform-pr.yml (excerpt)
- name: Run Infracost
  uses: infracost/actions/setup@v3
  with:
    api-key: ${ secrets.INFRACOST_API_KEY }

- name: Generate Infracost diff
  run: infracost diff --path infra/hub --compare-to infracost-base.json

- name: Post cost estimate to PR
  uses: infracost/actions/comment@v3
  with:
    path: /tmp/infracost.json
    behavior: update
```

This posts a cost delta comment on every PR, showing the cost impact of infrastructure changes.

5. AWS Billing Alerts

The following CloudWatch billing alarms will be created by Terraform:

Alarm	Threshold	Action
Monthly estimated charges	\$50	Email notification
Monthly estimated charges	\$100	Email notification
Monthly estimated charges	\$200	Email notification (warning)

```
# Billing alarms require CloudWatch in us-east-1 regardless of hub region
resource "aws_cloudwatch_metric_alarm" "billing_50" {
  alarm_name          = "aura-promptly-billing-50usd"
  comparison_operator = "GreaterThanThreshold"
  evaluation_periods  = 1
  metric_name         = "EstimatedCharges"
  namespace           = "AWS/Billing"
  period              = 86400
  statistic            = "Maximum"
  threshold            = 50
  alarm_actions       = [aws_sns_topic.billing_alerts.arn]
}
```

6. Cost Comparison: PoC vs Production

Aspect	PoC	Production (indicative)
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Aspect	PoC	Production (indicative)
LiteLLM compute	1 Fargate task (0.25 vCPU)	3–10 tasks (auto-scaling)
Aurora	Serverless, scale-to-zero, shared cluster	Provisioned (2 ACU min), Multi-AZ, dedicated cluster
Vector store	Aurora pgvector (shared cluster)	OpenSearch Serverless (4+ OCU) or dedicated pgvector
WAF	Basic rules	Advanced rules + Bot Control
Bedrock model	Haiku 3	Sonnet 3.5 or higher
Monthly cost	\$20–\$110 (toggleable)	\$2,000–\$10,000+

The PoC is intentionally 20–100x cheaper than production to enable learning without financial risk.